SOC LAB / TECHNICAL INVESTIGATION

SOC LAB PROJECT 01

Windows Failed Logon Detection and Investigation Using Wazuh

Technical case study from a controlled cybersecurity home lab • Public portfolio edition

Executive summary

This lab exercise demonstrates the recovery of a Windows endpoint monitoring configuration and the end-to-end collection of unsuccessful Windows authentication events. A Windows 11 virtual machine was connected to a Wazuh Manager hosted on Ubuntu Server. After verifying Windows logon auditing, an initial attempt to generate a failed-logon event using runas did not yield the expected Security event. A subsequent interactive sign-in test produced two Windows Event ID 4625 records, which were also visible as Wazuh alerts. Analysis of the event fields established that the account vboxuser was targeted and that Windows reported an incorrect password.

Scope: This case study follows the available lab evidence from endpoint monitoring through Windows auditing, controlled sign-in tests and Wazuh alert analysis. Background context is provided where necessary, and observed events are distinguished from interpretation.

1. Background: How the environment was established

Before the initial screenshot, the lab was configured in Oracle VirtualBox using two virtual machines: an Ubuntu Server running Wazuh Manager and a Windows 11 endpoint. An earlier Windows VM, FINANCE-PC-01, had been deleted and replaced by FINANCE-PC-02. This explains why checking the manager previously showed only the local server and why the Windows agent needed to be installed on the replacement VM.

The Ubuntu Manager was confirmed running, its local address was checked as 192.168.56.101, and a ping from the Windows VM demonstrated that the two machines could communicate. The Windows agent was then installed/configured on the replacement VM and the connection to the manager was verified. The screenshots in this report include overlapping confirmation of these steps.

2. Starting evidence: Wazuh endpoint dashboard

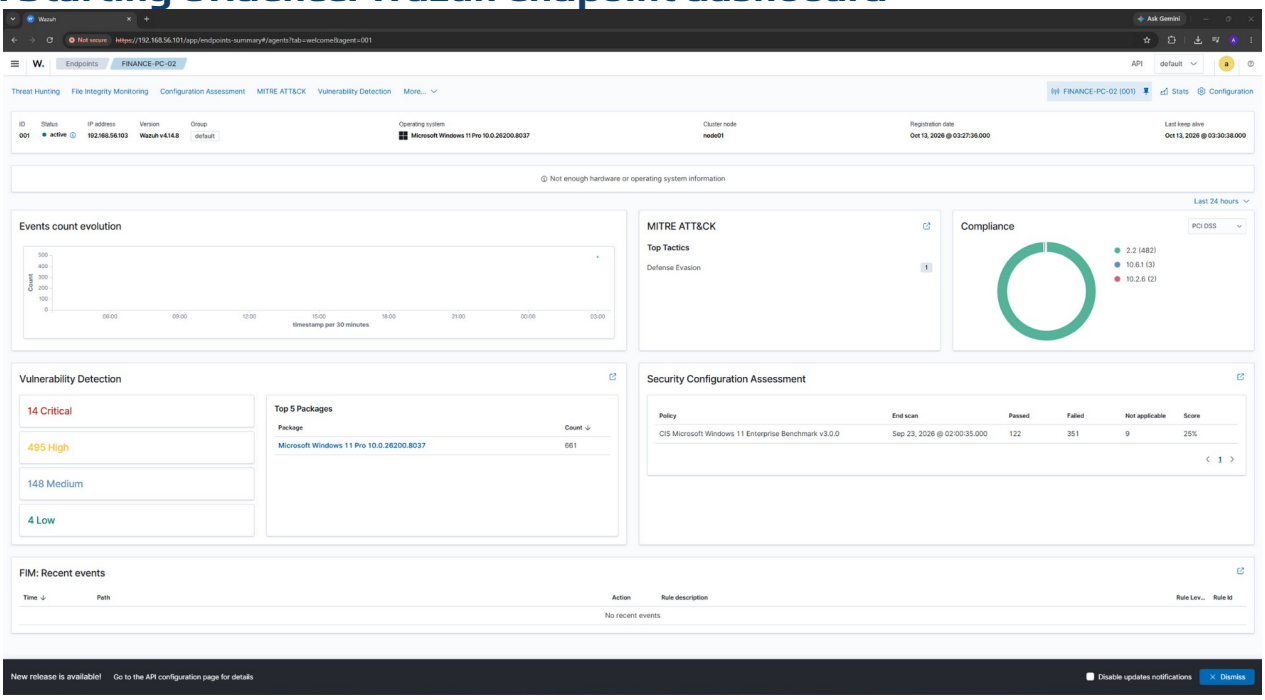


Figure 1. Wazuh endpoint dashboard showing FINANCE-PC-02 and endpoint security panels.

The first screenshot shows the Wazuh endpoint summary for FINANCE-PC-02. The agent is active and the dashboard displays its operating system, rule/event widgets, vulnerability information and security configuration assessment information. This gives a clear starting checkpoint: the monitored Windows endpoint was visible in the SIEM rather than being merely powered on in VirtualBox.

The dashboard also shows vulnerability counts and an assessment score. These are contextual dashboard findings, not the focus of this exercise; the evidence does not establish that any of those findings were exploited or remediated.

3. Installing and checking the Windows monitoring agent

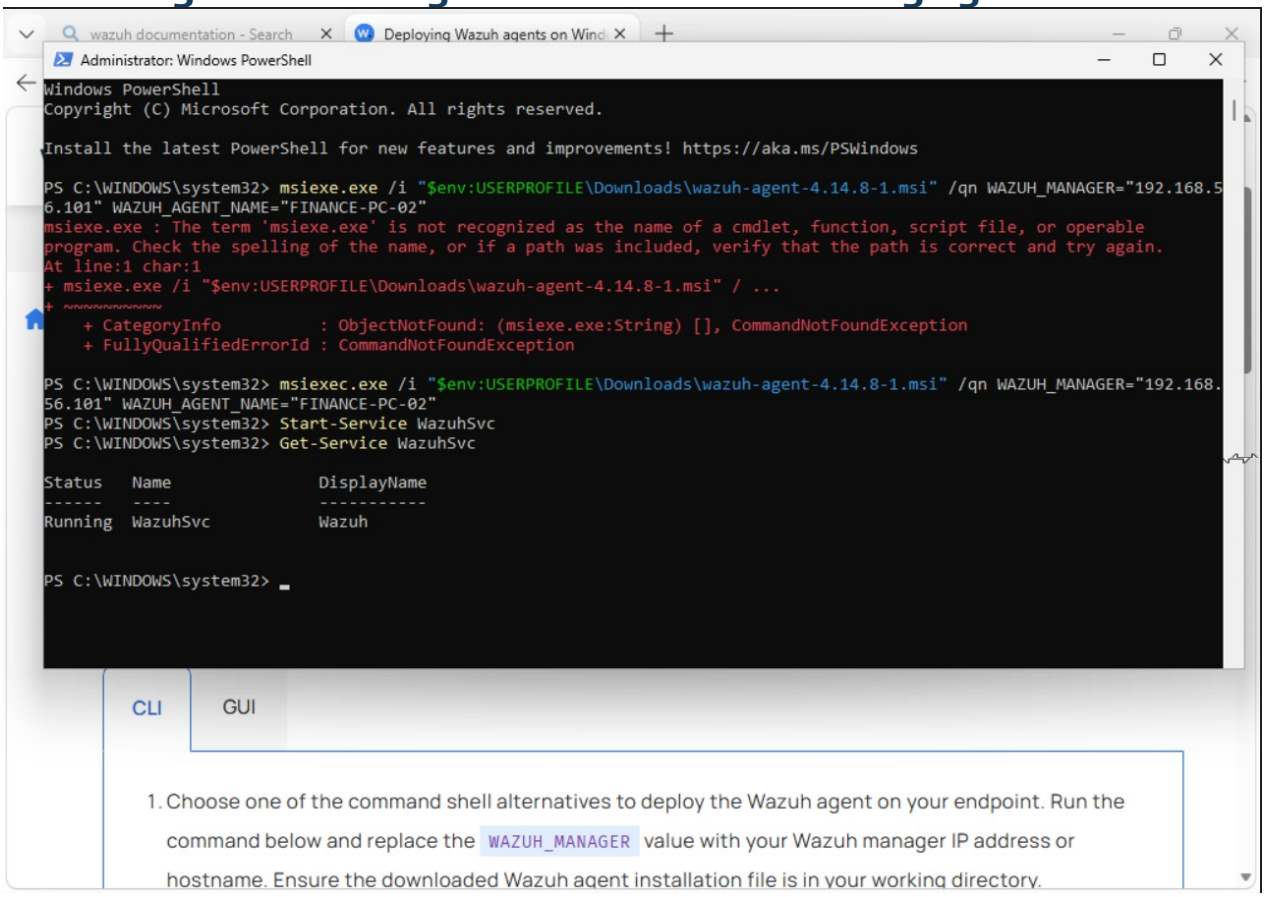


Figure 2. Windows Administrator PowerShell: unsuccessful mistyped installer command followed by successful service start and service-status check.

The Windows screenshots show an initial command-entry error: msiexe.exe was not recognised by PowerShell. The intended executable was msiexec.exe, the Windows Installer command-line program. After correcting the spelling, the installer was run with parameters specifying the Wazuh Manager address and the endpoint name FINANCE-PC-02.

```
msiexec.exe /i "$env:USERPROFILE\Downloads\wazuh-agent-4.14.8-1.msi" /qn WAZUH_MANAGER="192.168.56.101" WAZUH_AGENT_NAME="FINANCE-PC-02"

Start-Service WazuhSvc
Get-Service WazuhSvc
```

The service query returned Running for WazuhSvc. This establishes that the Wazuh agent service started on the Windows guest; the stronger confirmation of successful enrolment is the manager-side registration shown next.

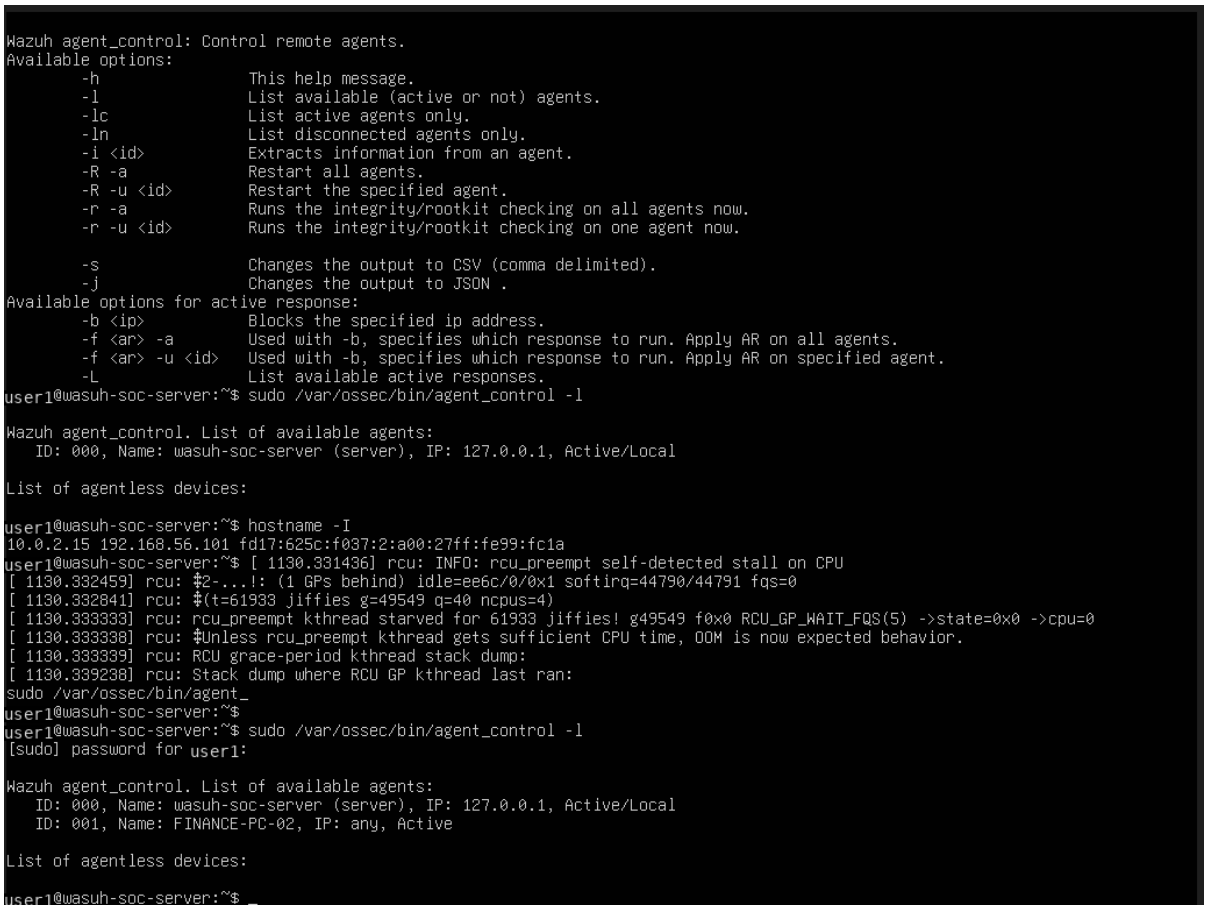


Figure 3. Ubuntu agent-control output before and after registration: initially only the local manager, subsequently FINANCE-PC-02 listed as active.

On Ubuntu, the command below was used to inspect registered agents. The captured terminal evidence includes a point at which only ID 000 (the local Wazuh server) was available and a later check listing ID 001 for FINANCE-PC-02 as active. The change confirms agent registration at manager level, not just a locally running Windows service.

```
sudo /var/ossec/bin/agent_control -l
```

4. Confirming Windows logon audit configuration

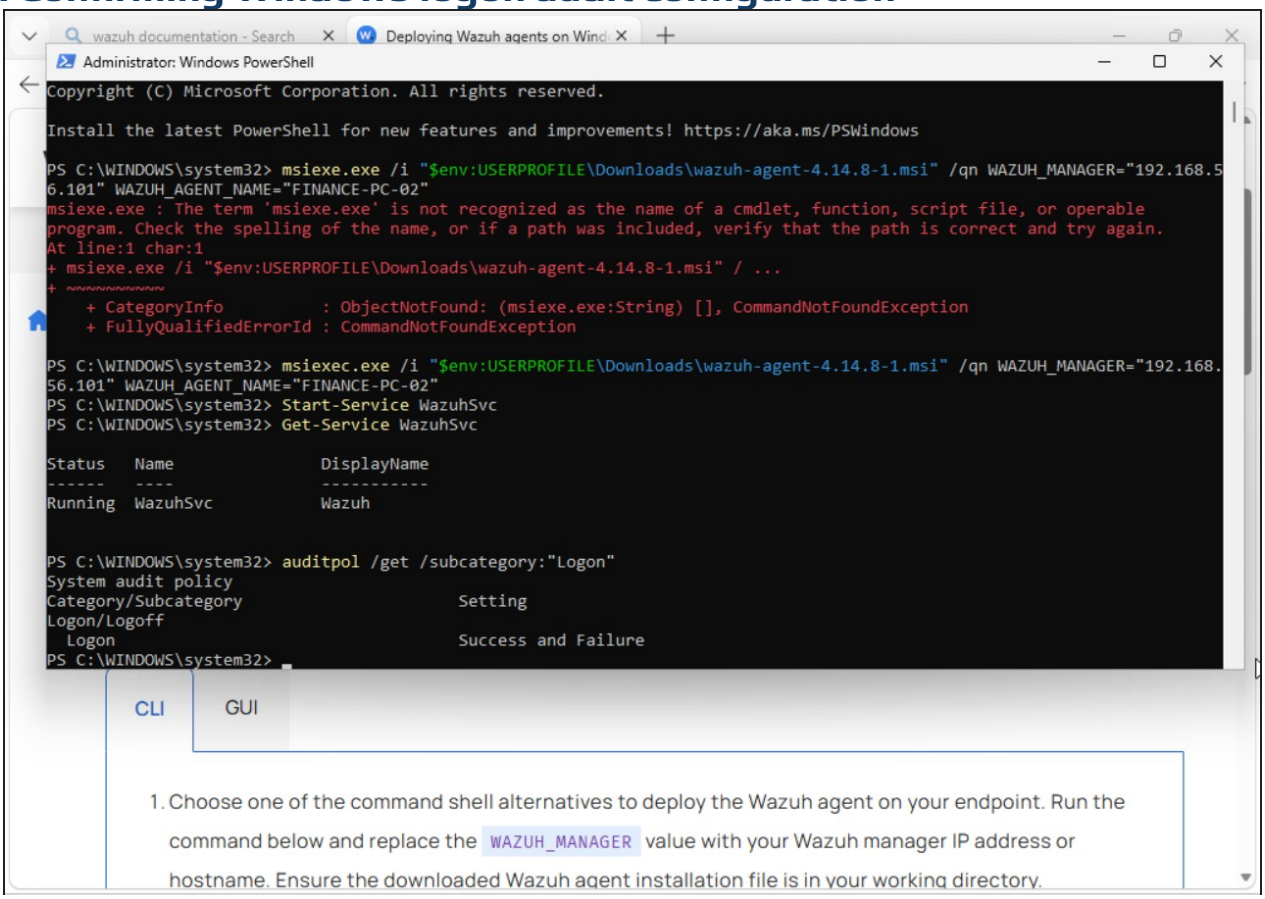


Figure 4. Windows PowerShell: WazuhSvc running; logon audit policy reports Success and Failure.

To ensure that Windows could produce the desired security telemetry, the Logon audit subcategory was queried using auditpol. The output was Success and Failure. This is an essential prerequisite for investigating normal Security log event 4625: without auditing, failed authentication attempts may not be recorded in the expected Windows Security channel.

```
auditpol /get /subcategory:"Logon"
```

This check verified an existing setting; the screenshots do not show that the audit policy needed to be changed.

5. Initial attempt: simulated failure with runas

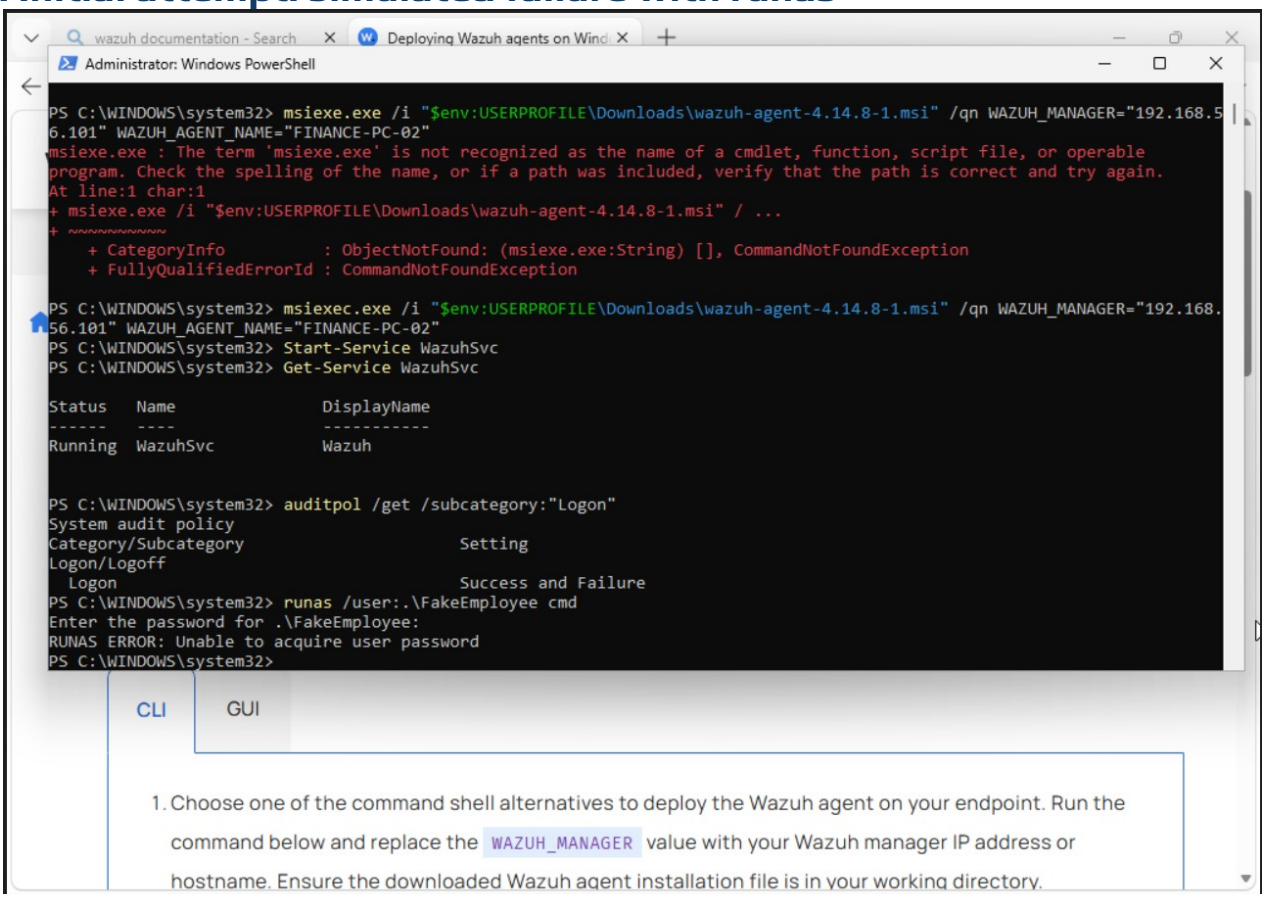


Figure 5. PowerShell command preparing a runas test using an intentionally nonexistent local account.

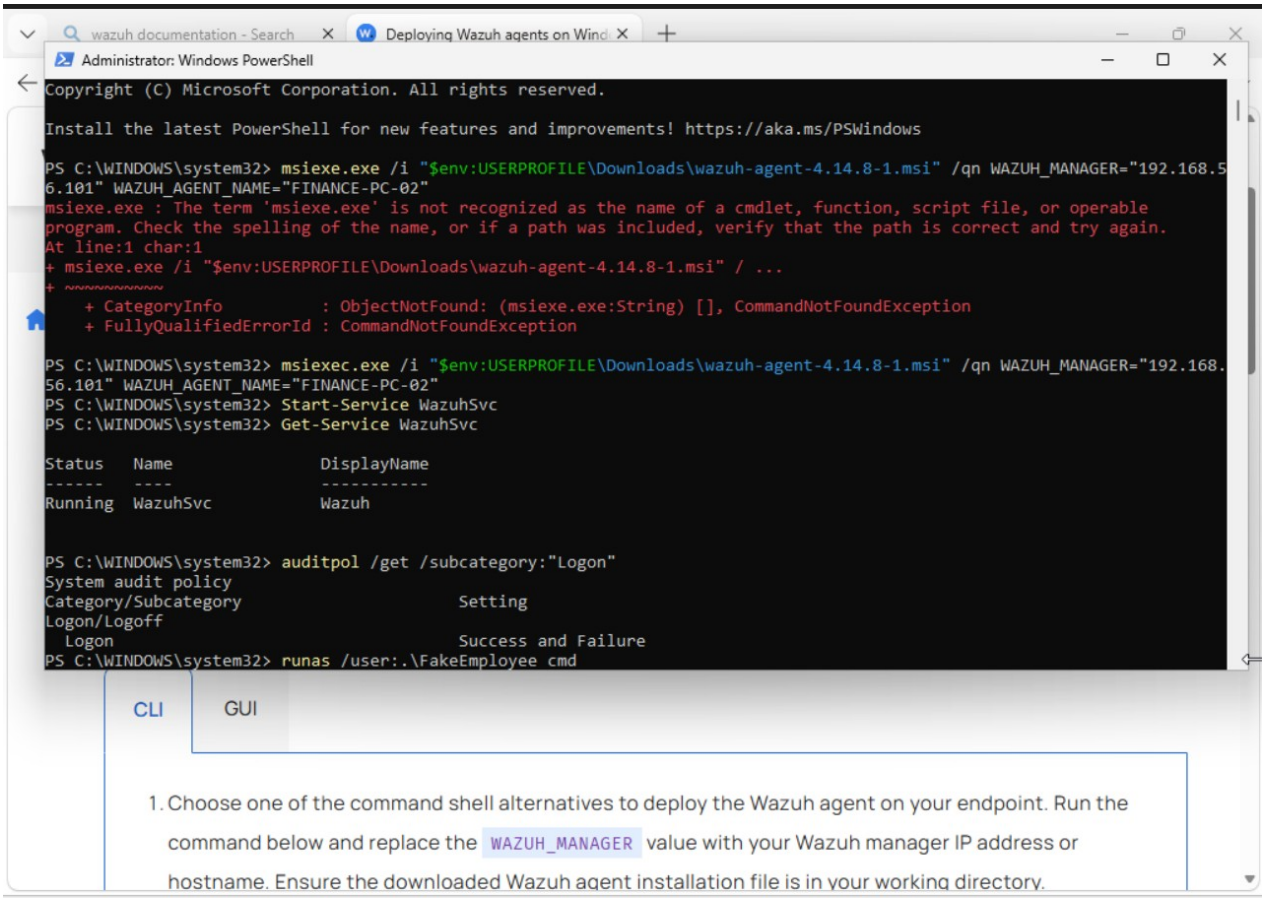


Figure 6. runas prompted for a password and returned an error instead of launching the requested process.

The first test attempted to launch a command shell as a nonexistent local user, FakeEmployee, using the runas utility. Windows requested a password and returned RUNAS ERROR: Unable to acquire user password. Although this was an unsuccessful authentication-related action, it was not sufficient evidence that Windows had generated the particular audit event required for the investigation.

```
runas /user:.\FakeEmployee cmd
```

This distinction is significant: a visible command failure is not the same thing as a logged Event ID 4625. The next checks therefore looked directly for the event before making any claim about Wazuh detection.

6. Checking the Windows Security event log

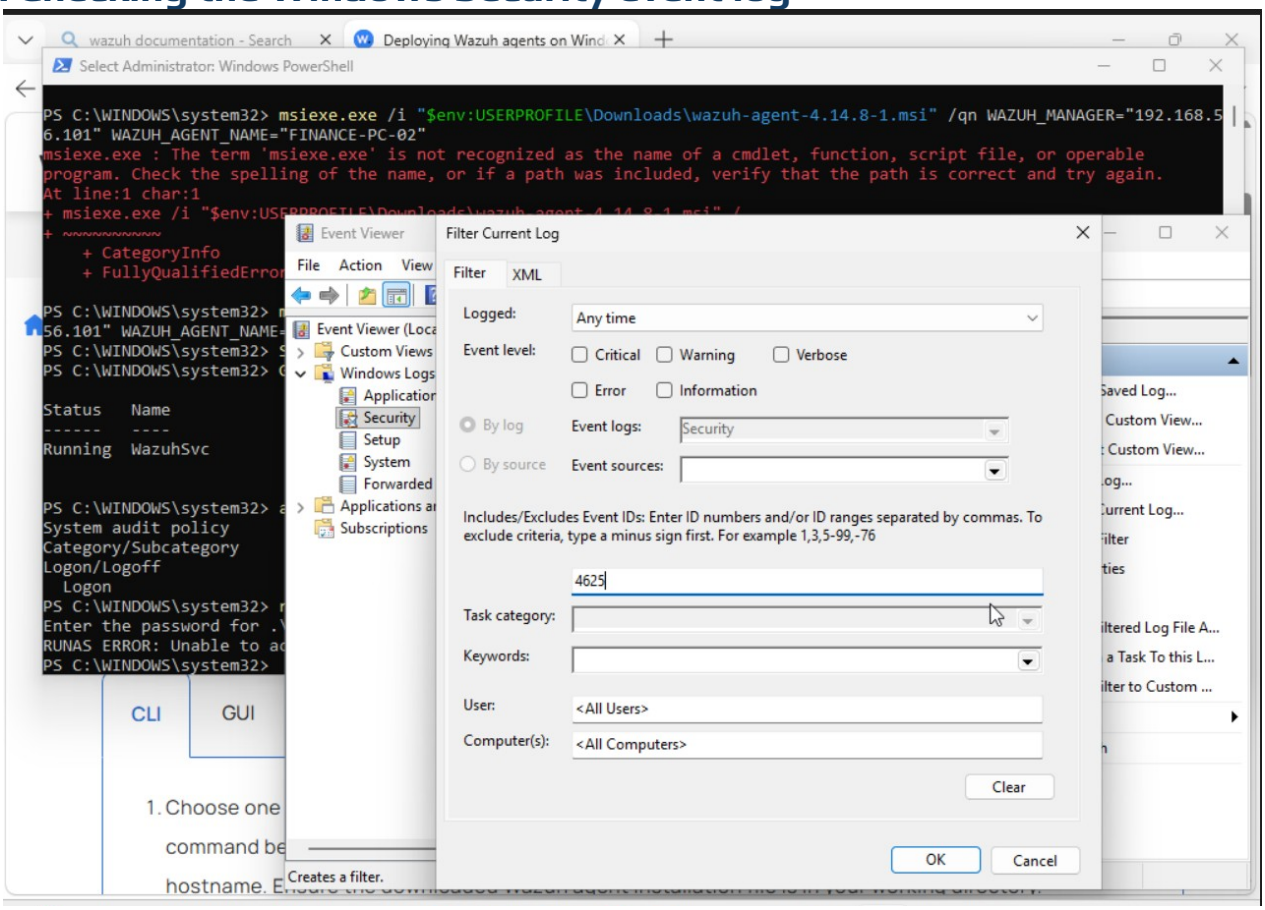


Figure 7. Event Viewer: Windows Logs > Security, filtered for Event ID 4625.

Event Viewer was opened and the Security log filtered for Event ID 4625. The filtered view initially displayed no matching events. This did not prove that Wazuh was malfunctioning; it indicated that the event had not yet been confirmed at the source.

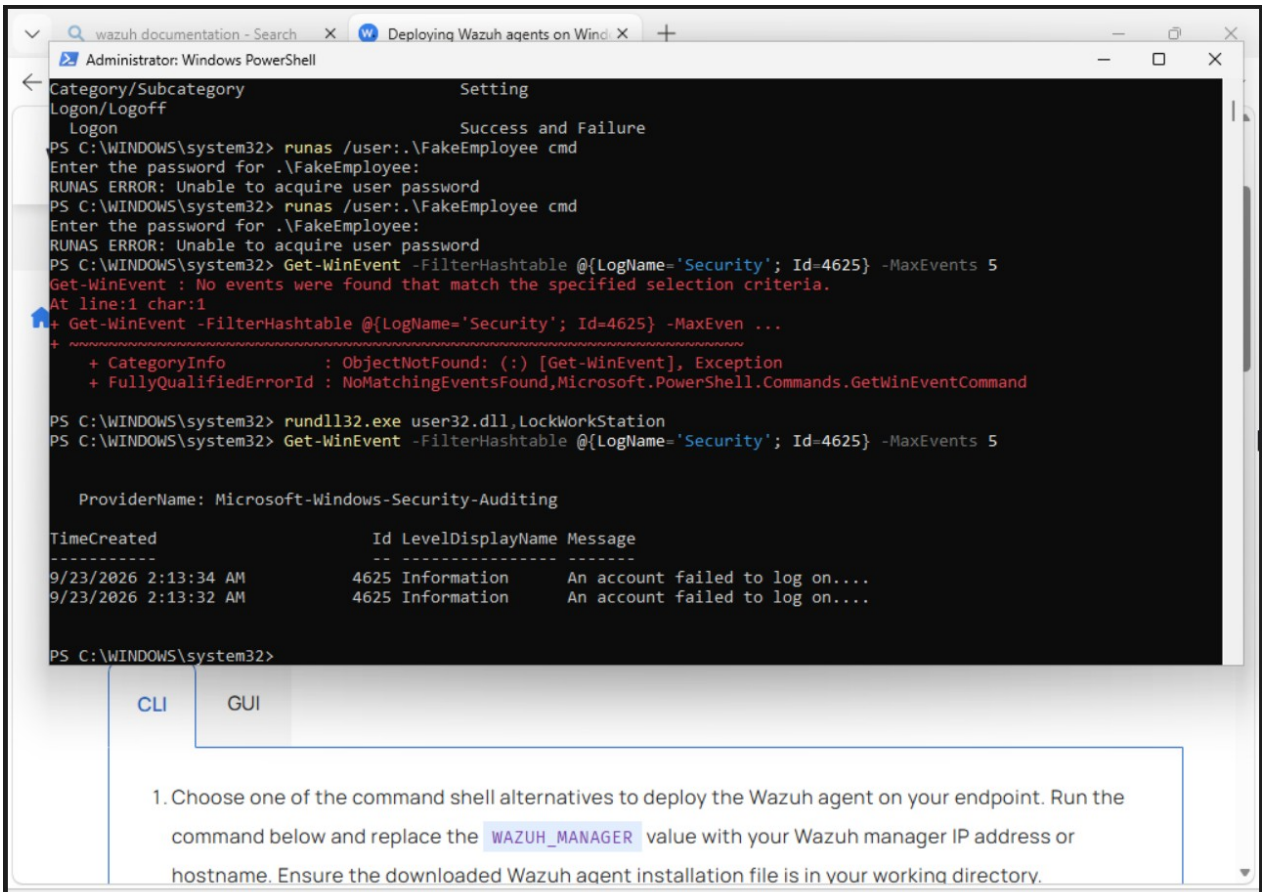


Figure 8. PowerShell verification: an initial query returned no matching events; after a new interactive sign-in test, two Event ID 4625 records were found.

Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 5

The same test was repeated through PowerShell. The initial response was “No events were found that match the specified selection criteria.” A different test was then conducted: the Windows workstation was locked and deliberately incorrect credentials were entered at the sign-in screen, after which the user signed back in.

rundll32.exe user32.dll,LockWorkStation

Repeating the Get-WinEvent query now returned two entries, each with Event ID 4625 and the message “An account failed to log on.” This established the local source evidence for the controlled sign-in failure. Note that the screenshot includes Windows VM dates in September; later dashboard captures show October. These captures should be treated as evidence of the same investigated setup/workflow, not as proof that every screenshot was recorded in one uninterrupted session.

7. Confirming the alerts in the Wazuh SIEM

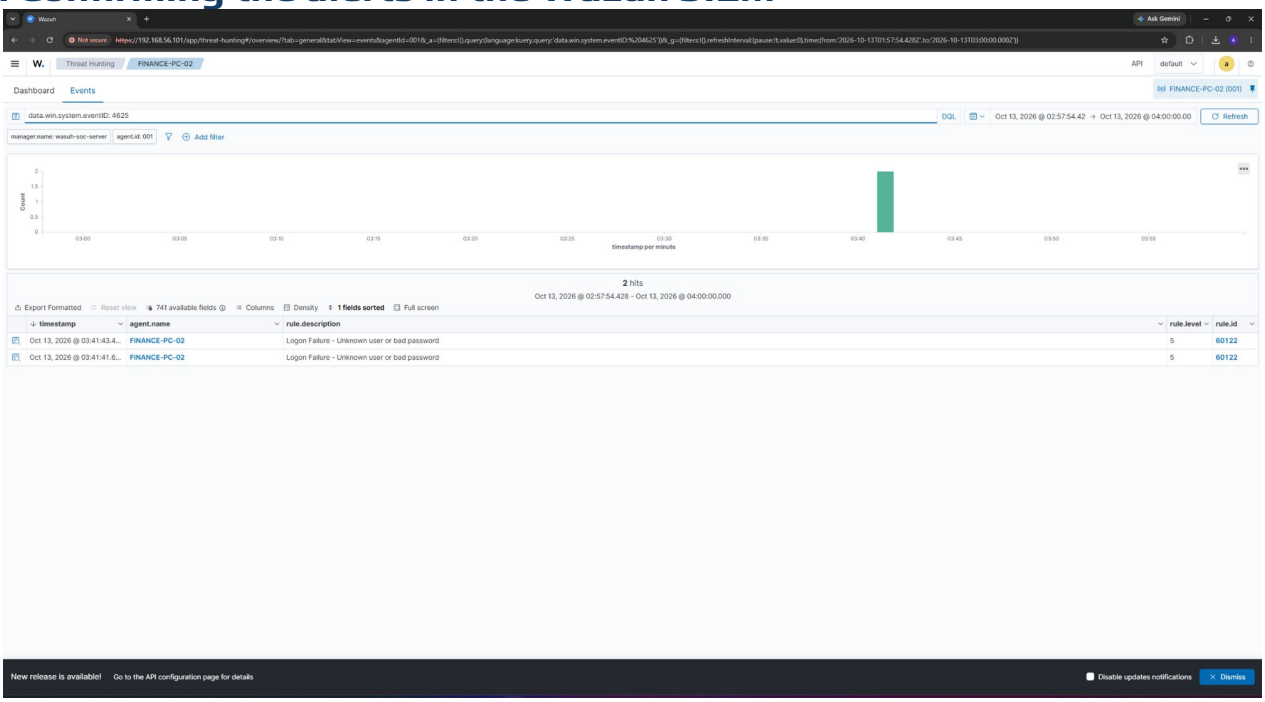


Figure 9. Wazuh Threat Hunting: search for Windows Event ID 4625 returns two alerts from FINANCE-PC-02.

The investigation moved from the endpoint to Wazuh Threat Hunting. Filtering on data.win.system.eventID: 4625 while scoped to agent 001 returned two hits. Both entries were associated with FINANCE-PC-02 and displayed the description “Logon Failure - Unknown user or bad password,” Wazuh rule ID 60122 and rule level 5. The screenshot therefore links the Windows security events to their ingestion and categorisation in the SIEM.

data.win.system.eventID: 4625

Two such alerts in this controlled context are not evidence of brute-force activity or compromise. The alert count and rule level make them reviewable, but the test itself explains the observed activity.

8. Inspecting the alert details

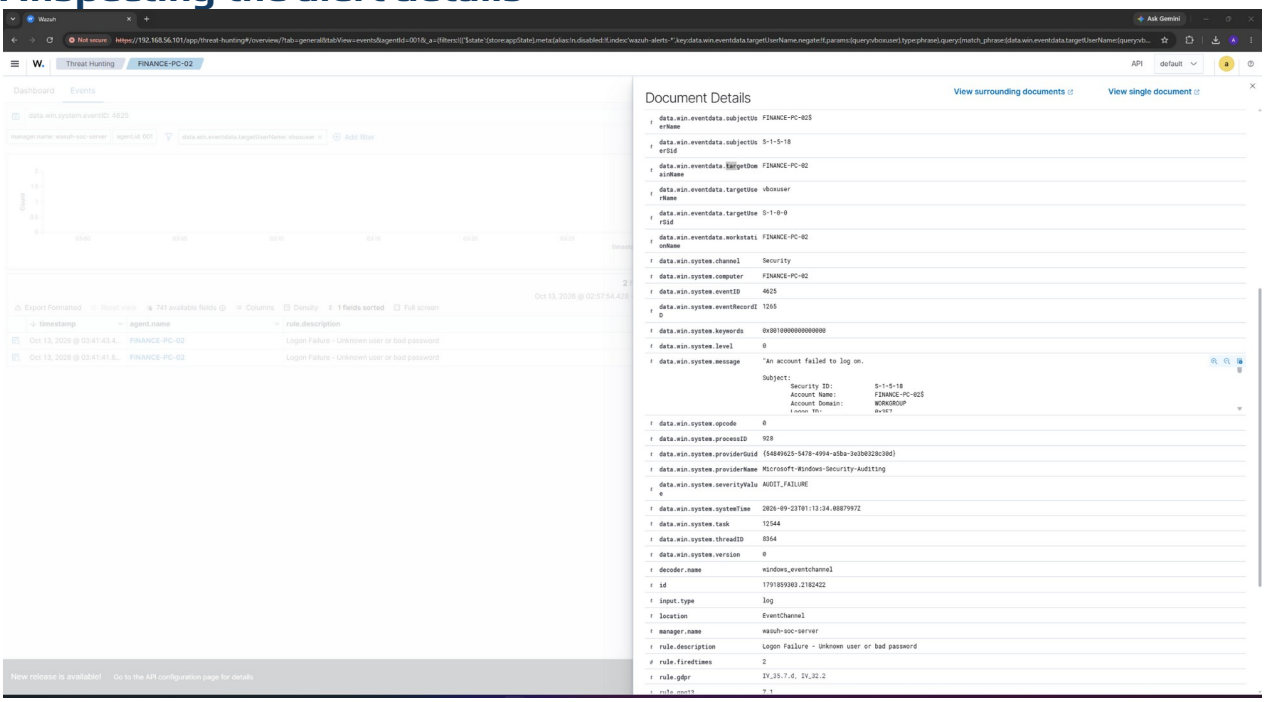


Figure 10. First expanded Wazuh document view showing Windows eventdata fields and the recorded 4625 event.

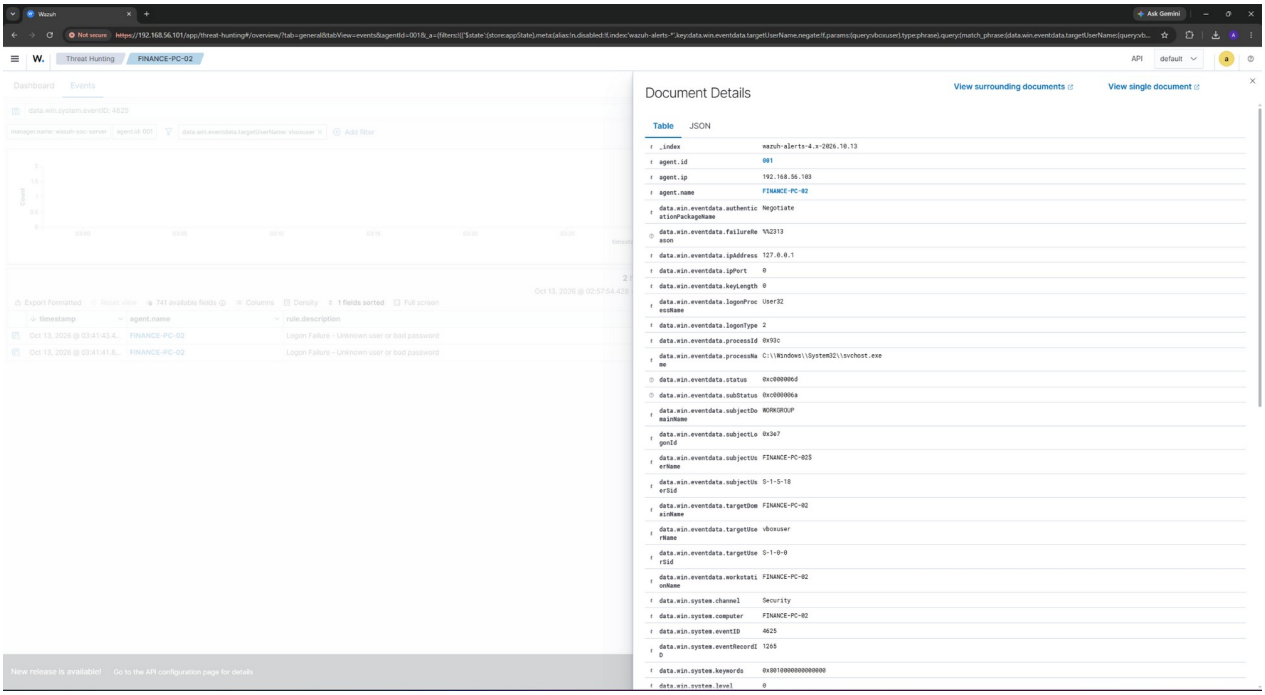


Figure 11. Additional Wazuh document detail fields supporting authentication analysis.

The expanded documents exposed the underlying Windows audit fields. The target account was vboxuser on FINANCE-PC-02. The event was categorised as an audit failure. The following interpretations are drawn from the detailed event screenshot also inspected during this investigation:

Evidence	Interpretation
Event ID 4625	Windows recorded an unsuccessful logon

	attempt.
targetUserName = vboxuser	The username the sign-in attempt tried to authenticate.
status = 0xC000006D	General authentication failure.
subStatus = 0xC000006A	The supplied password was incorrect.
logonType = 2	Interactive logon, consistent with a local sign-in test.
ipAddress = 127.0.0.1	Loopback/local address, not evidence of a remote external source.
processName = C:\Windows\System32\svchost.exe	A Windows service-host process named in the record; not, by itself, evidence of malware.
Wazuh rule 60122, level 5	SIEM rule matched the failed logon and classified it for review.

The event also included workstation name FINANCE-PC-02 and local Windows Security auditing metadata. The failed password itself was not recorded; Windows audit events expose authentication outcomes and related metadata rather than plaintext credentials.

9. Timeline and chain of evidence

Evidence	Interpretation
Endpoint and manager ready	Wazuh dashboard showed an active Windows agent, while manager-side agent_control demonstrated active registration.
Logon auditing confirmed	auditpol reported that successful and unsuccessful logons were audited.
Initial test inconclusive	runas returned an error; Event Viewer and Get-WinEvent found no 4625 records at that stage.
Interactive test successful	After deliberate incorrect local sign-ins, Get-WinEvent returned two 4625 audit records.
SIEM correlation successful	Wazuh Threat Hunting returned two corresponding alerts attributed to FINANCE-PC-02.
Cause interpreted	Logon type 2 and substatus 0xC000006A supported an incorrect-password explanation.

10. Results, limitations and lessons learned

Outcome: The exercise successfully demonstrated that a controlled failed interactive Windows login can be recorded in the Security log, collected by a Wazuh Windows agent, forwarded to the Ubuntu-based manager and examined centrally in the Wazuh Threat Hunting interface. The strongest evidence consists of the matching two-event count, event ID 4625, agent attribution and detailed authentication status fields.

Limitations: This was a controlled lab test, not a genuine intrusion; it did not establish brute-force activity, successful compromise, alert escalation or incident remediation. The screenshots also span different date displays, and the first runas failure did not generate a confirmed 4625 record. No custom detection rule was developed in the documented work.

Technical lessons: (1) Confirm basic connectivity and service state before troubleshooting event collection. (2) Validate audit-policy prerequisites. (3) Check whether an event exists on the source system before blaming the SIEM. (4) Use status and substatus codes rather than generic alert descriptions to identify the precise failure. (5) Preserve screenshots and commands, including unsuccessful attempts, as reproducible investigation evidence.

Conclusion

The lab achieved its defined objective: reconstructing a deliberate Windows failed-logon event across both the source endpoint and the Wazuh SIEM. It offers an introductory, evidence-based demonstration of endpoint monitoring, basic Windows authentication auditing, SIEM searching and event triage. A logical subsequent project would examine repeated failed logons, alert thresholds and incident documentation; those activities are proposed future work, not claimed as completed here.

Evidence note: All screenshots were captured in a personal virtual lab. For this public version, the Ubuntu account username in the terminal screenshot was anonymised; remaining hostnames and private lab addresses are retained to explain the investigation.